Data Visualization using Tableau

Session 3

Agenda

- Advanced Dashboard Techniques in Tableau

Mastering Calculated Fields

Calculated Fields

Calculated fields in Tableau are custom fields created by using formulas or expressions to transform existing data or generate new data points. They allow users to perform complex calculations, derive new metrics, and manipulate data directly within Tableau without altering the original data source.

- **Custom Formulas**: Utilize Tableau's rich formula language to perform calculations, including arithmetic operations, logical conditions, and text manipulations.
- **Dynamic Calculation**: Calculated fields are recalculated automatically as the data context changes, such as when filters are applied, or dimensions are adjusted.
- **Versatile Use**: Can be used in various places, including rows, columns, filters, marks, tooltips, and more.
- **Enhance Analysis**: Enable deeper insights by deriving new metrics, combining data fields, and creating conditional outputs.

Mastering Calculated Fields

Types of Calculated Fields

1. **Basic Arithmetic Calculations**:
   - **Example**: Calculate total cost as **'Cost' * 'Quantity'**.
   - **Usage**: Creating new numerical measures like total expenses, discounts, or growth rates.'

2. **String Manipulations**:
   - **Example**: Combine **'First Name'** and **'Last Name'** into a full name using **'First Name' + ' ' + 'Last Name'**.
   - **Usage**: Formatting and concatenating text fields, extracting parts of strings.

3. **Date Calculations**:
   - **Example**: Calculate the age of a record using **DATEDIFF('year', [Birth Date], TODAY())**.
   - **Usage**: Working with dates to calculate durations, age, or future/past dates.

Mastering Calculated Fields

Types of Calculated Fields

4. **Logical Calculations**:
   - **Example**: Create a field to categorize sales as **’IF [Sales] > 10000 THEN 'High' ELSE 'Low' END’**.
   - **Usage**: Conditional statements to create flags, segments, or categories based on data conditions.

5. **Aggregate Calculations**:
   - **Example**: Calculate average sales per customer using **’SUM([Sales]) / COUNTD([Customer ID])’**.
   - **Usage**: Aggregating data to find averages, totals, or distinct counts.

6. **Level of Detail (LOD) Expressions**:
   - **Example**: Calculate the average sales per region regardless of the view's filters using **’FIXED [Region] : AVG([Sales])’**.
   - **Usage**: Controlling the level of aggregation to create measures at different levels of detail.

رواد مصر الرقمية

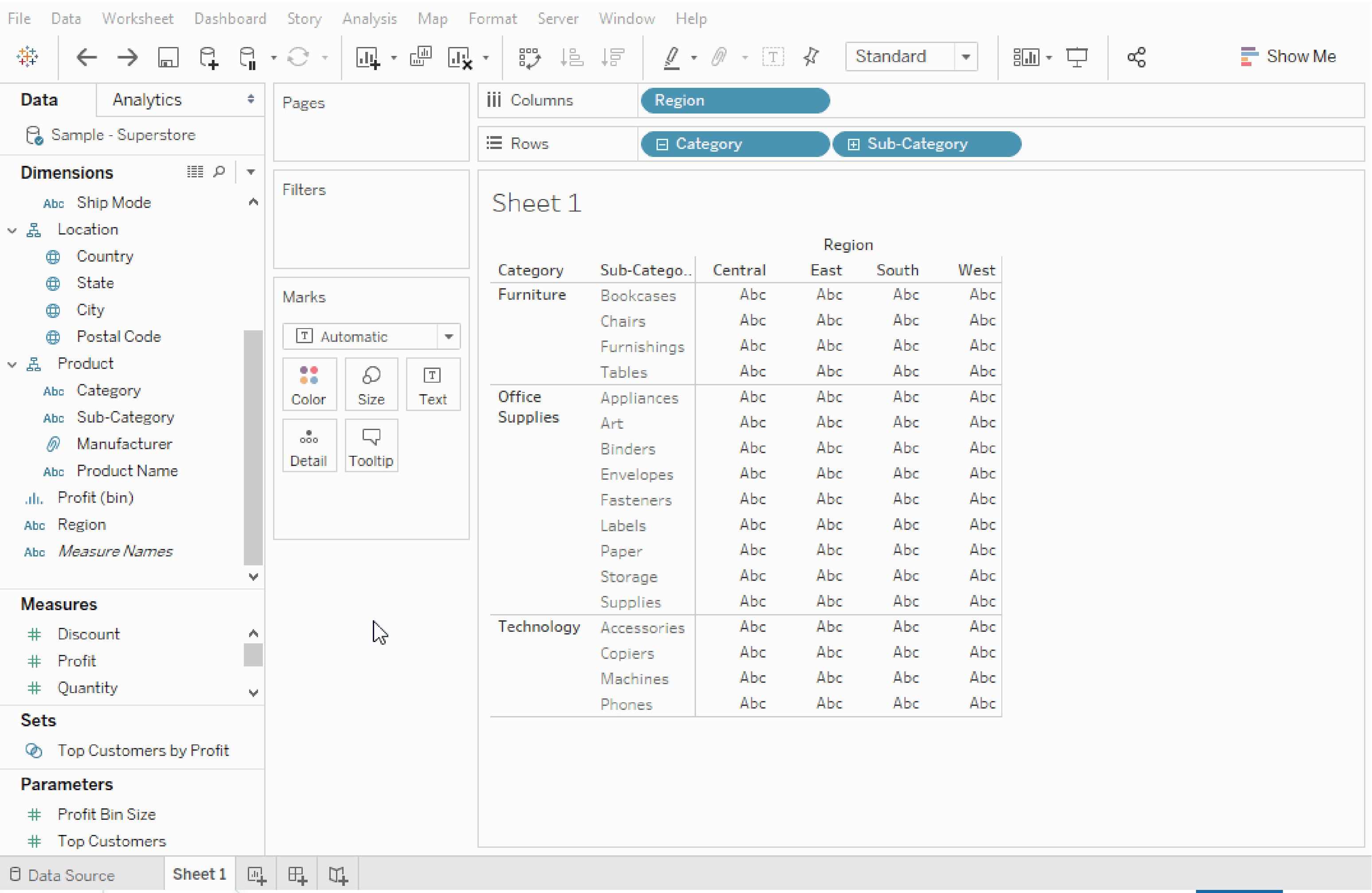
Mastering Calculated Fields

وزارة الاتصالات وتكنولوجيا المعلومات

Examples:

- **Profitability Index:**
- (SUM([Sales]) - SUM([Cost])) / SUM([Cost])

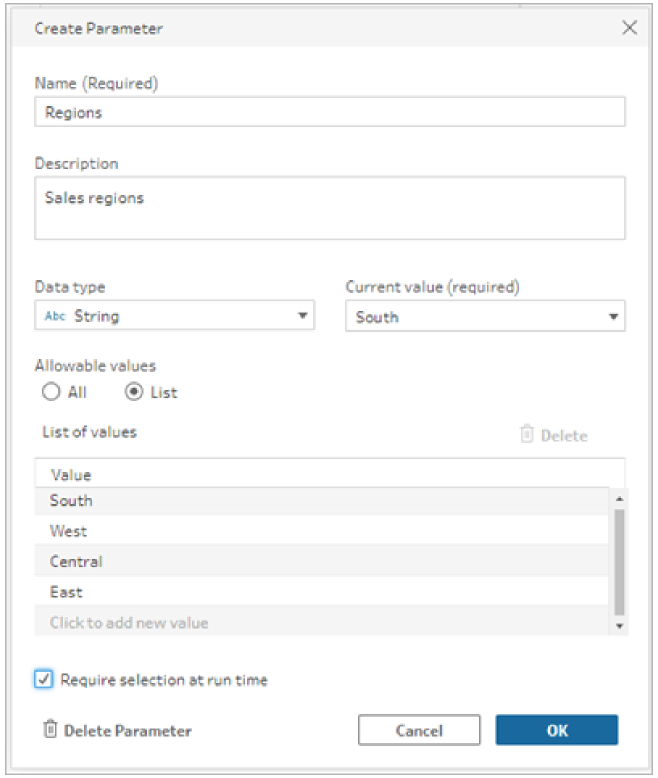
- **Customer Engagement Duration:** DATEDIFF('month’
- , MIN([First Purchase Date])
- ,MAX([Last Purchase Date]))



Utilizing Parameters

Parameters

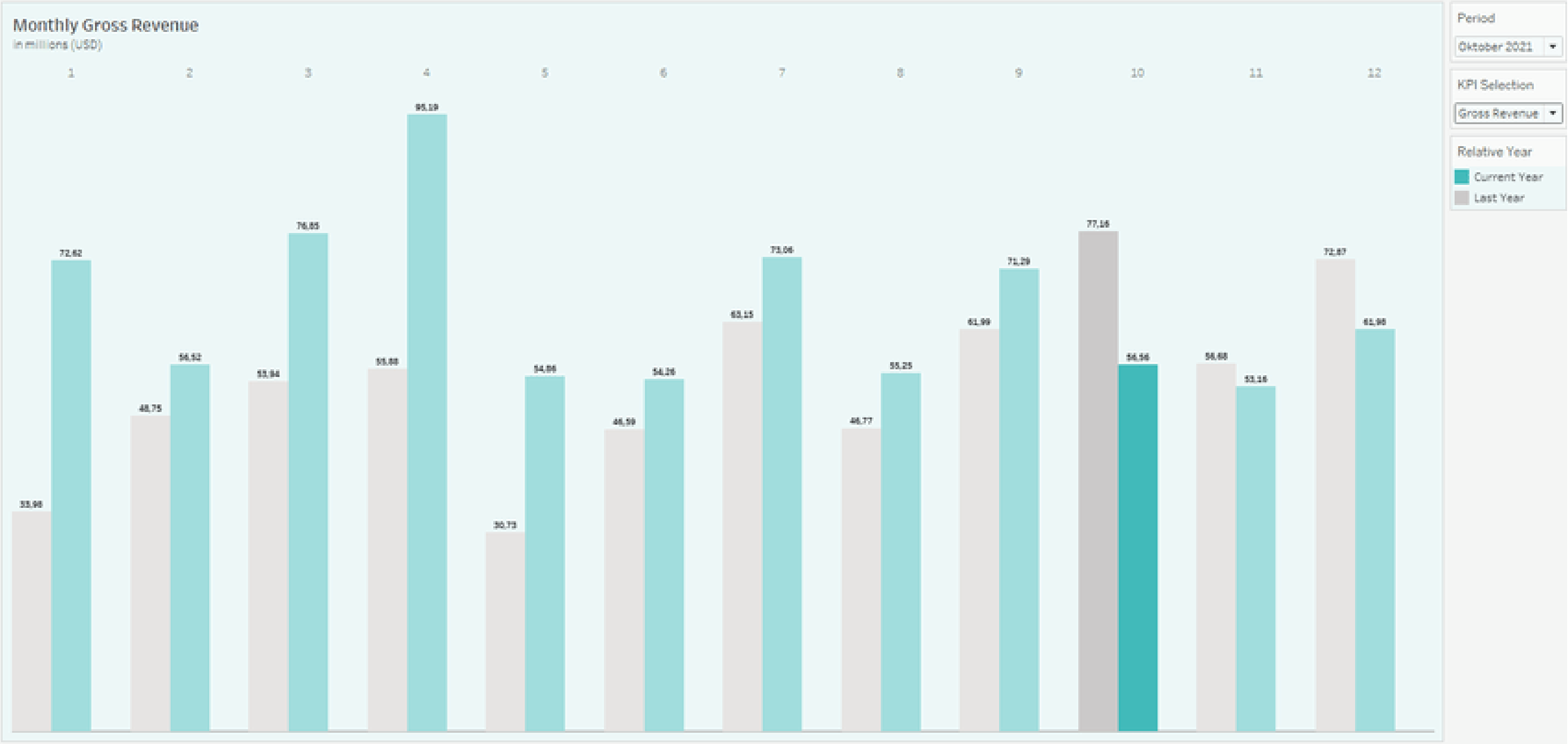
A parameter is a global placeholder value such as a number, text value, or Boolean value that can replace a constant value in a flow.
Instead of building and maintaining multiple flows, you can now build one flow and use parameters to run the flow with your different data sets.



Utilizing Parameters

Purpose of Parameters: Parameters act as placeholders that users can control to manipulate data views dynamically.

Case Study Example: A dashboard that allows users to select a region or period, updating visualizations such as sales data and market trends.



Filter Actions

Filter Actions in Tableau allow users to create interactive dashboards where selecting a data point in one view dynamically filters data in other related views. This interaction enhances the dashboard experience by providing context-sensitive data exploration and seamless navigation between different data perspectives.

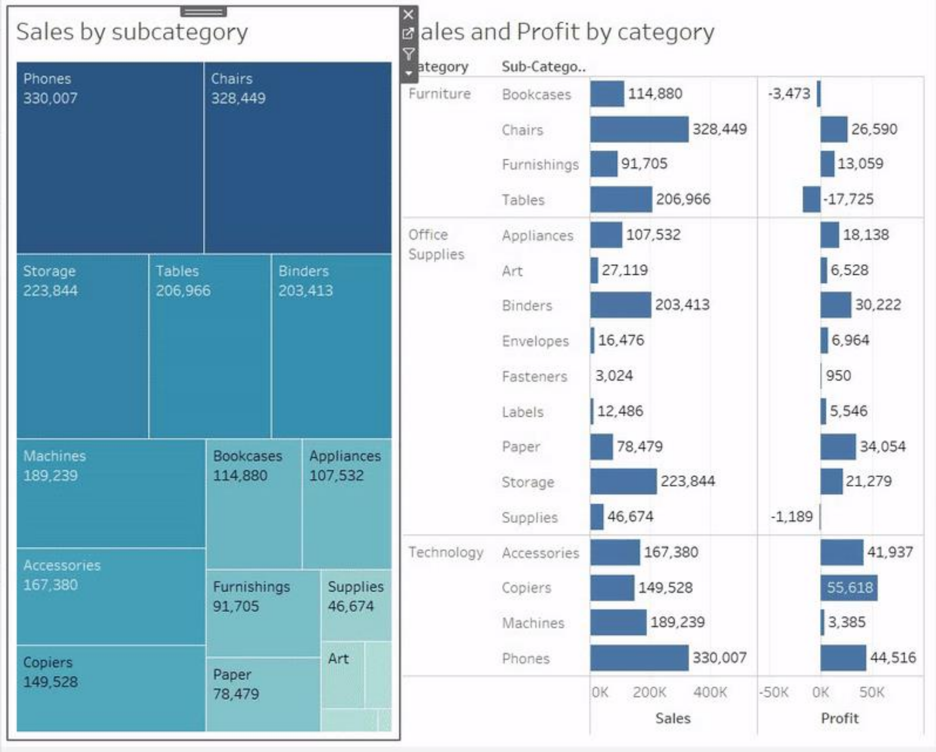
- **Interactivity**: Users can click on a specific part of one visualization to filter data displayed in another.
- **Contextual Filtering**: Only data relevant to the selected item(s) is shown in connected views.
- **Flexible Control**: Filter actions can be customized to control which fields and views are affected.
- **Intuitive Navigation**: Provides a natural and intuitive way for users to drill down into details or change the focus of analysis within a dashboard.

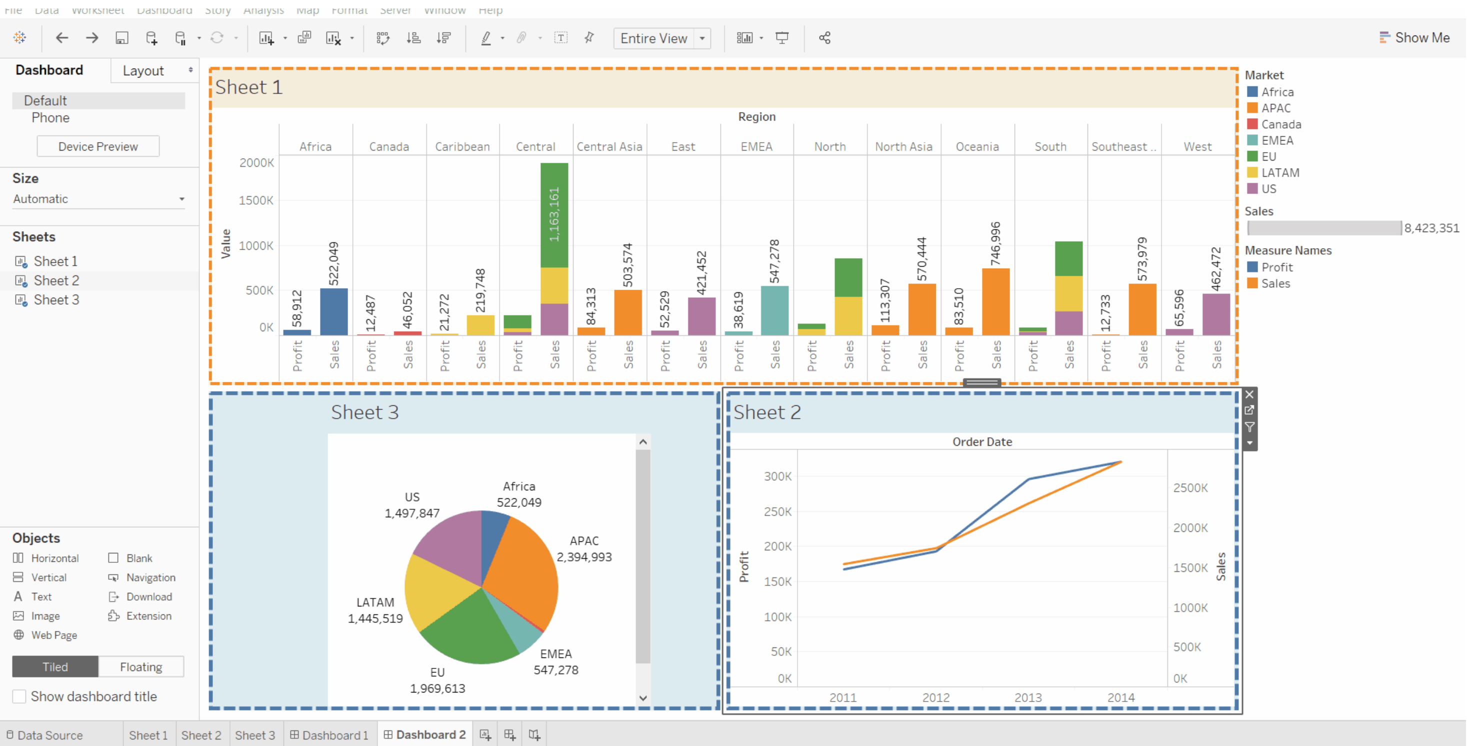
رواد مصر الرقمية

Advanced Filter Actions

وزارة الاتصالات وتكنولوجيا المعلومات

Functionality: Filter actions provide a powerful way to make dashboards interactive.
Practical Application: Set up a dashboard where selecting a product category automatically updates associated metrics like sales trends and profitability.
Benefits: Streamlined data exploration and a cohesive user experience across multiple data points.





رواد مصر الرقمية

Let's practice

وزارة الاتصـــــالات
وتكنولوجيا المعلومات

Warm up exercise - dynamic time grouping (10 Minutes)

- The idea is to show how to create an option to switch between different ways of grouping dates (years / months / quarters, etc.)
- create a new worksheet name it "Grouped date"
- Add a new parameter named Order Date Group select string values as data type and allowable values as list and start typing: year, month, quarter, week
- Show parameter
- Add a new calculated field named Order Date Grouped (it will use a different function from the ones that we have seen before):

```
if [Order Date Group] = 'quarter' then datename('year', [Order Date]) + '-Q' + datename('quarter', [Order Date])
elseif [Order Date Group] = 'year' then datename('year', [Order Date])
elseif [Order Date Group] = 'month' then datename('year', [Order Date]) + '-' + datename('month', [Order Date])
else datename('year', [Order Date]) + '-w' + datename('week', [Order Date])
end
```

- Add sales to rows
- Add order date grouped to columns & change it to dimension
- See how the chart changes when choosing different values for the parameter

Tableau Filters

- **Extract Filter:** limit the data before coming into Tableau.
- **Data Source filter:** data is loaded to tableau but filtered before going into worksheets.
- **Context filters:** Context filters create a temporary table in the Tableau engine and act as a primary filter before other Dimension and Measure filters are applied.
- **Dimension and Measure:** Filters will be executed after the context filter.

Tableau Filters

- Drag The Sales to column and Subcategory to rows.
- Add subcategory to filters pane and choose top 4
- Right click on Category > click Show filter
- Select Office Supplies

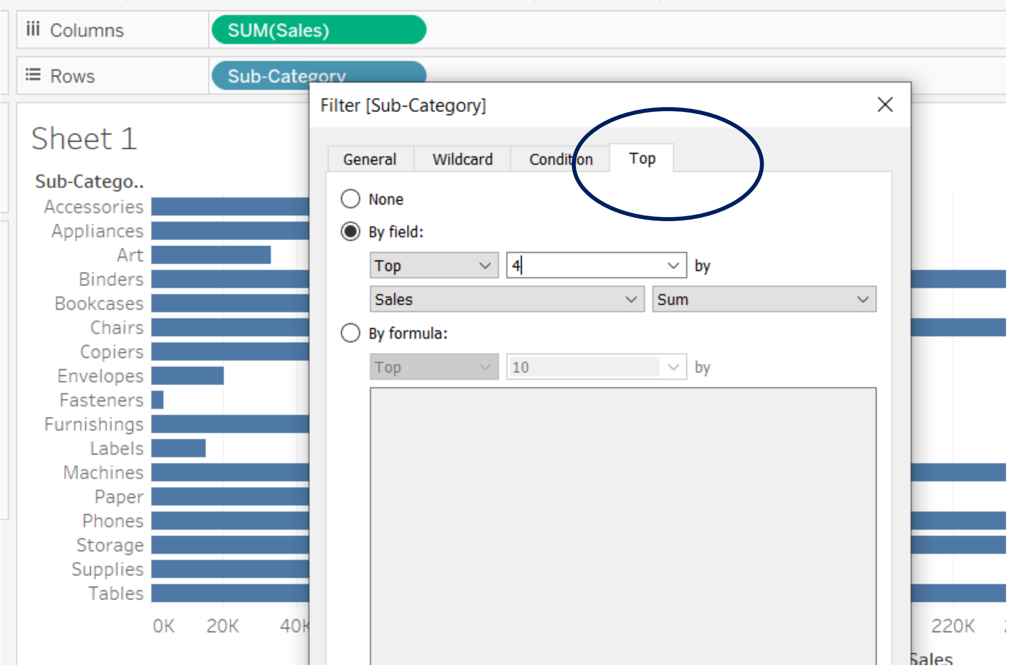


Tableau Filters

- Now the view is filtered, but instead of showing the top 4 Office Supplies products, it shows only one.
- This is because by default the dimension filters are executed separately and the view shows the intersection of the results.
- So, in this case what happens is that for the top 4 calculation, Tableau takes all rows in the dataset and calculates the top 4.
- For the Office Supplies filter, it once again looks at all rows in the data and keeps only Office Supplies items.
- Then it will look how many of the top 4 overall items are also Office Supplies items.

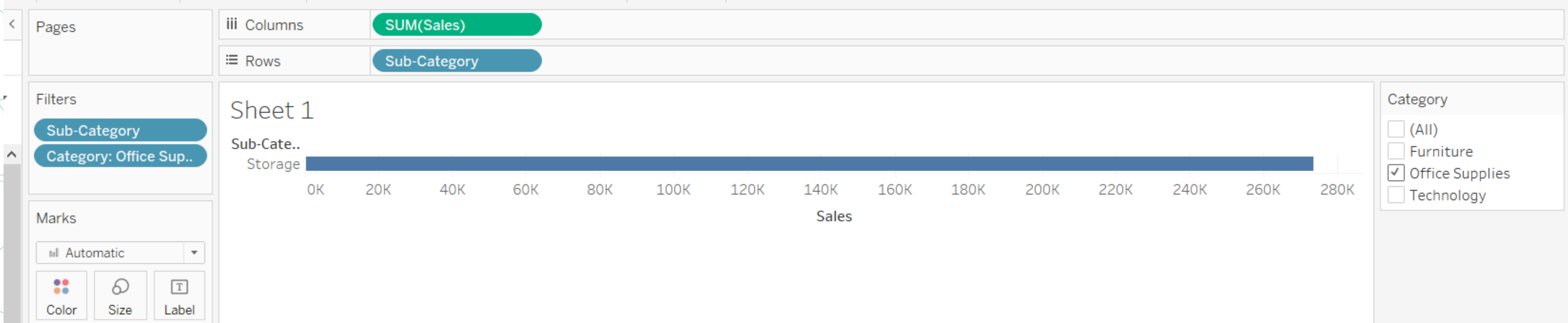


Tableau Filters

- Right Click on the category filter and click add to context.
- Now it shows the top 4 selling products in the office supplies category.
- Because of Tableau's order of operations, Context filters will be executed before Dimension and Measure filters.

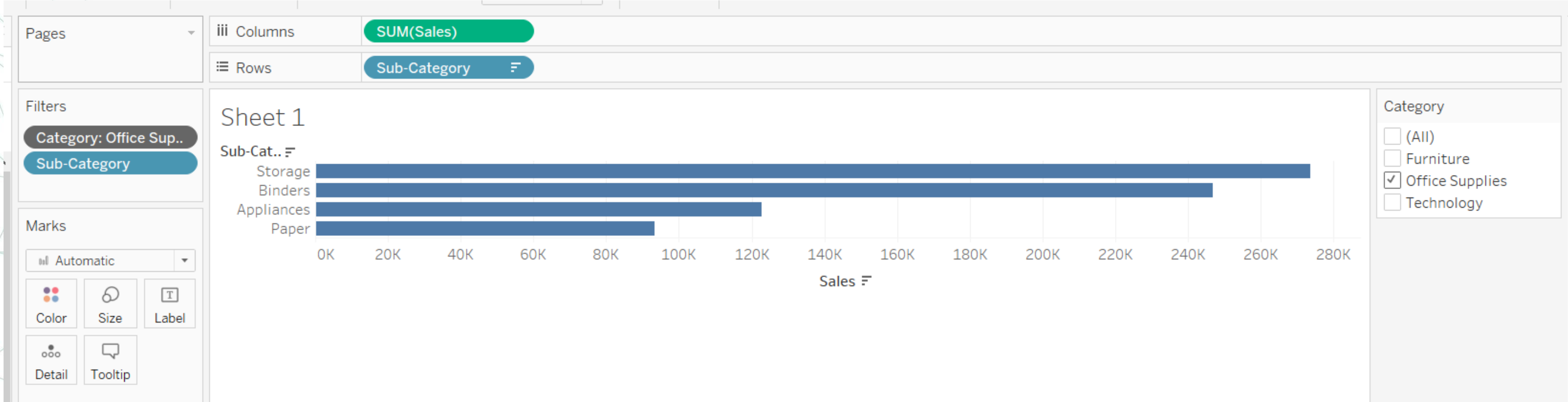


Tableau Filters

Filters can be applied to one or multiple worksheets , data source ,...

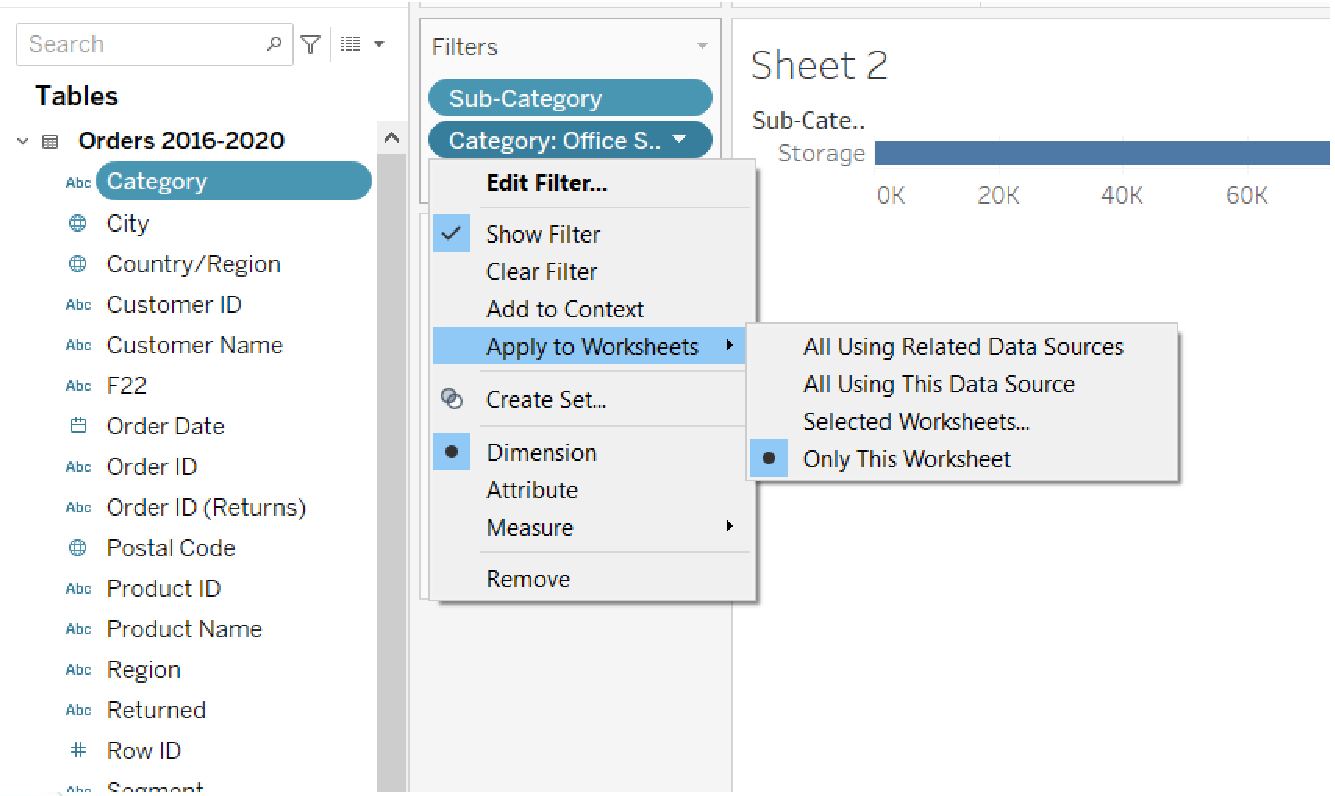
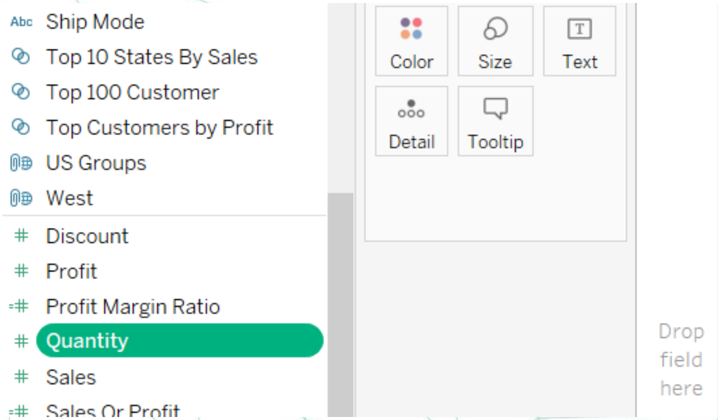


Tableau Bins (Self Practice 15 Minutes)

- Just like a bar plot, you plot a histogram by creating bars that contain the counts of the number of observations.
- Right-click the continuous measure of interest (Quantity) , then Create, then Bins.
- We can take a value of one here to begin, since it makes sense to look at the amount of orders where one item was ordered, two items were ordered, and so on. So change 1.77 to 1.
- A new field will be created Quantity (bins) , move it columns and Quantity to Rows.



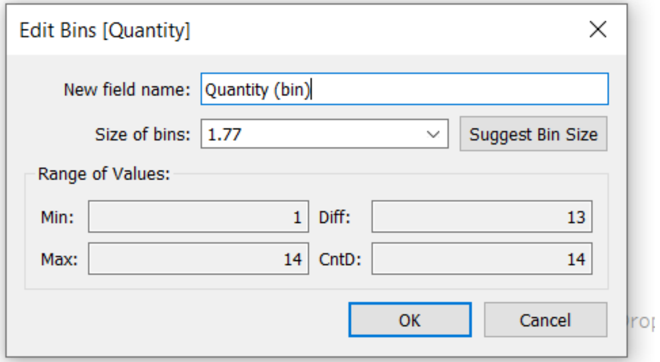


Tableau Bins (Self Practice)

- Change sum(quantity) to count.
- By default, bins are treated as discrete fields. This means that only the starting value of each bin is displayed. You might want to change that, especially when you are working with larger bin sizes.
- Right-click the bin field on the Columns shelf, and select Continuous.

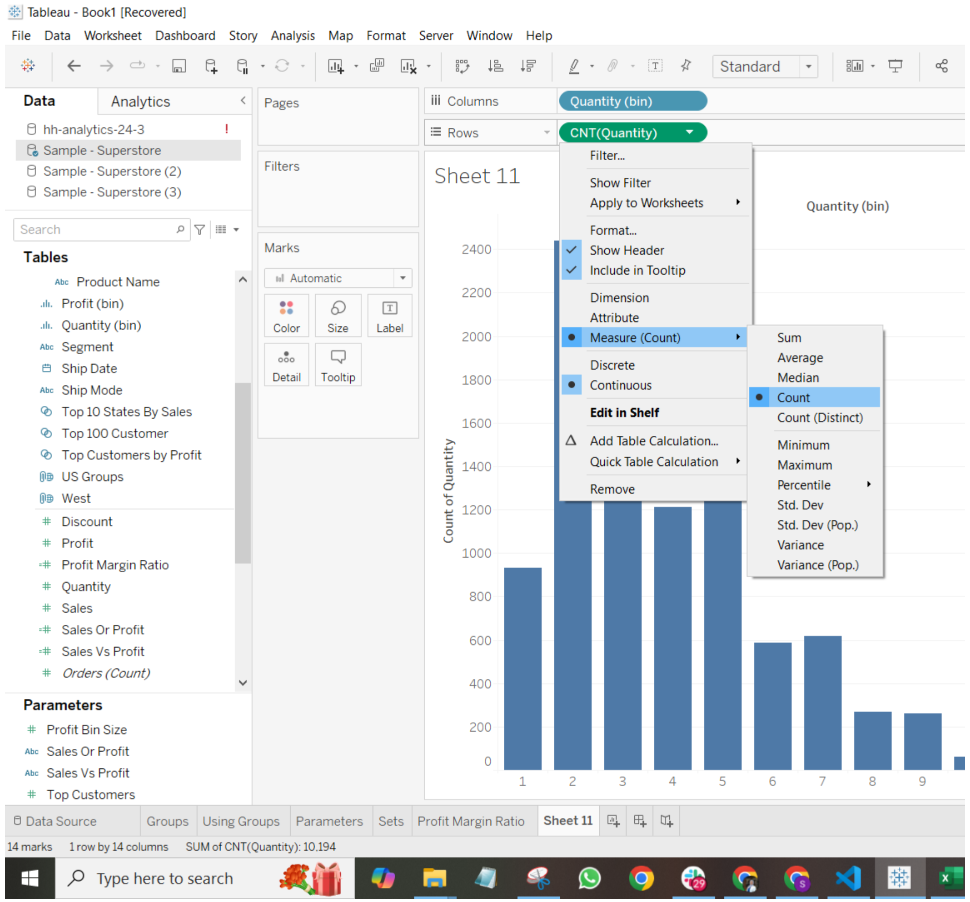


Tableau Bins (Self Practice)

- There is also a way of creating histograms in Tableau with just three clicks: first, you select the numerical measure of interest, then you click the 'Show Me' button, then the histogram view.

- Behind the scenes, Tableau repeated the steps we did manually: creating bins, aggregating by count and displaying it as a bar plot.

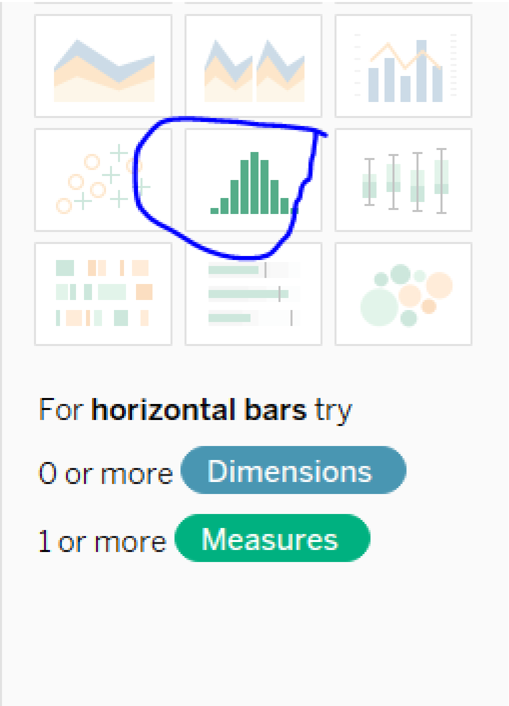
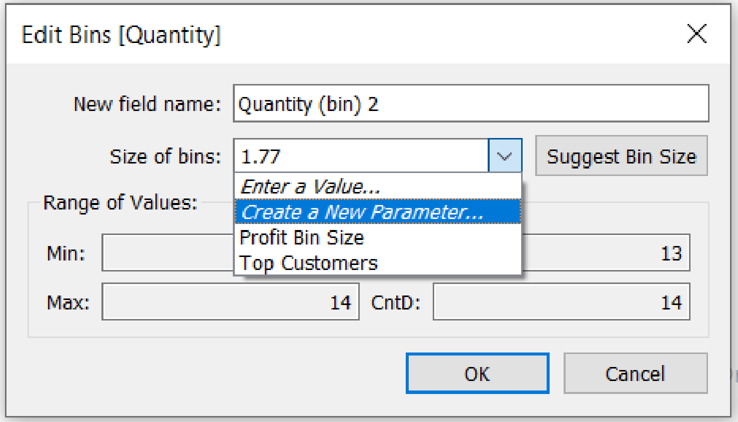
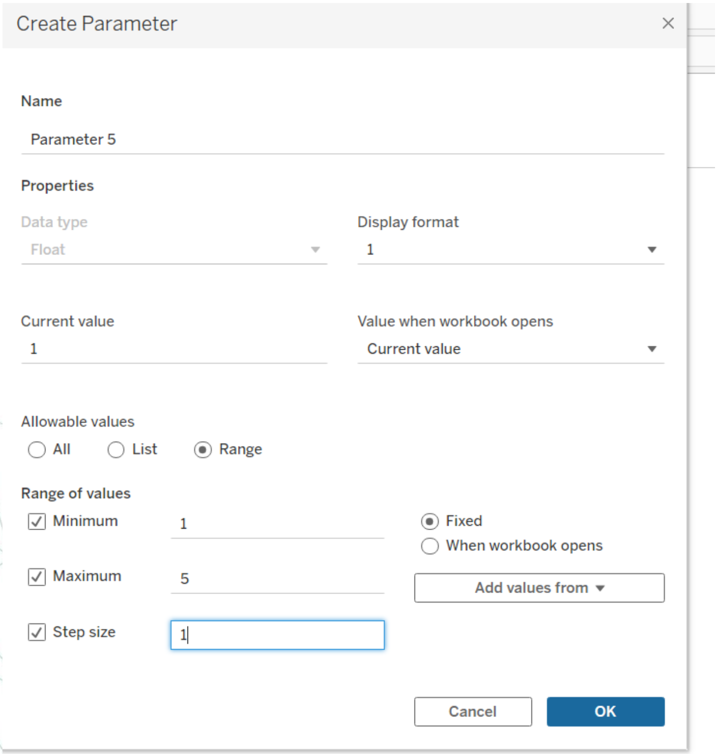


Tableau Bins & Parameters(Self Practice)

- A useful trick for EDA with histograms in Tableau, is setting a bin size using a Parameter.
- By choosing 'Create a New Parameter' instead of a constant value, you'll be able to specify a range of values in which the bin size should be tested.
- Click Show Parameter and change it to see the effect of the histogram.





LODs Expressions in Tableau

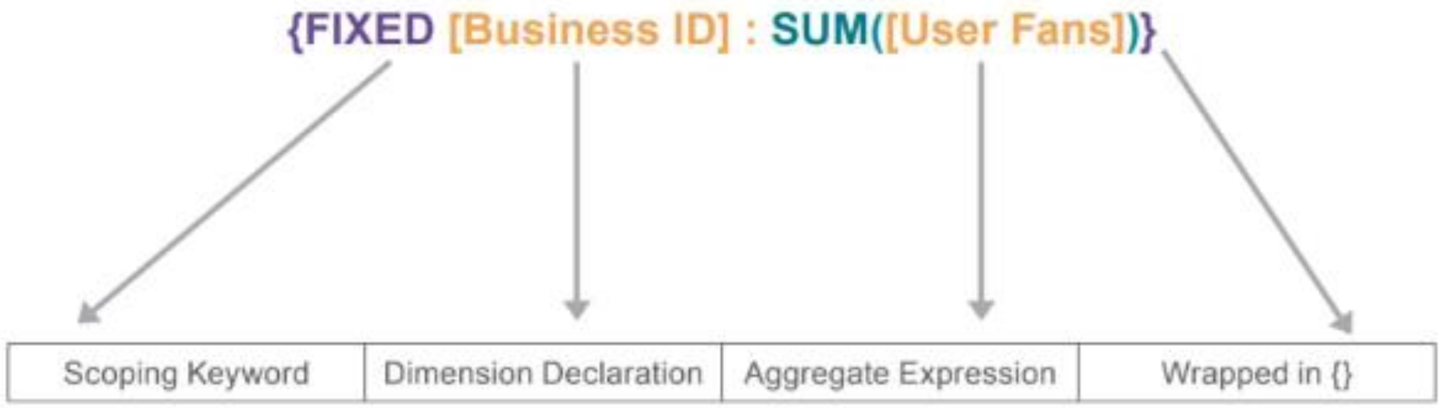
- Enter LOD expressions - Tableau's elegant and powerful way to easily compute aggregations that are not at the level of detail of the visualization.

- There are 3 functions in this family: INCLUDE, EXCLUDE, and FIXED.

- Through these functions you control the level of detail at which a calculation is performed and do not depend on dimensions used in the visualization. We will first look at the FIXED expression.

Fixed LOD

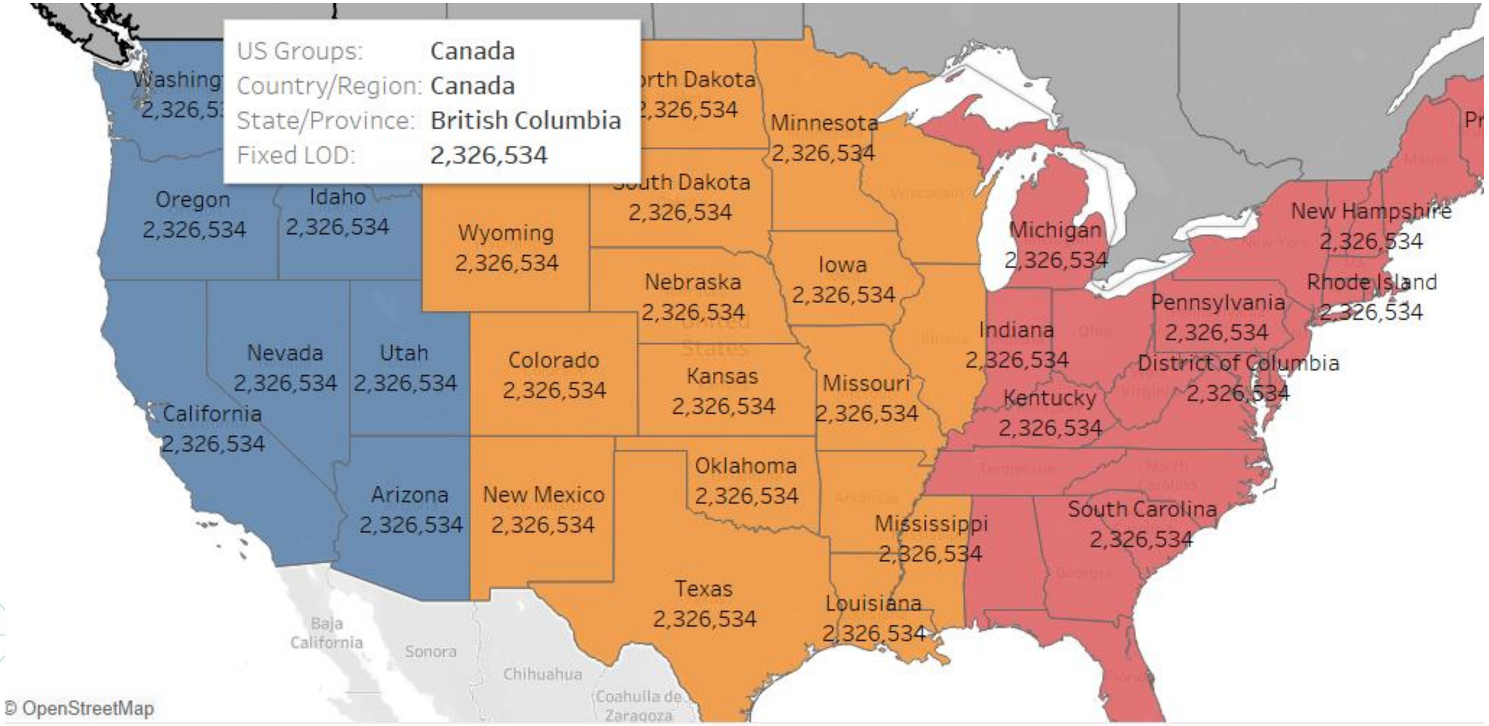
FIXED LOD expressions compute a value using the specified dimensions, without reference to the dimensions in the view.



Fixed LOD

- Generate The map again.
- Create a calculated field.
- Name it 'fixed grand total sales’.
- Use {fixed: sum([Sales])}
- Add it to label.

You can see that the resulted total sales on each state instead of the default aggregation per state.



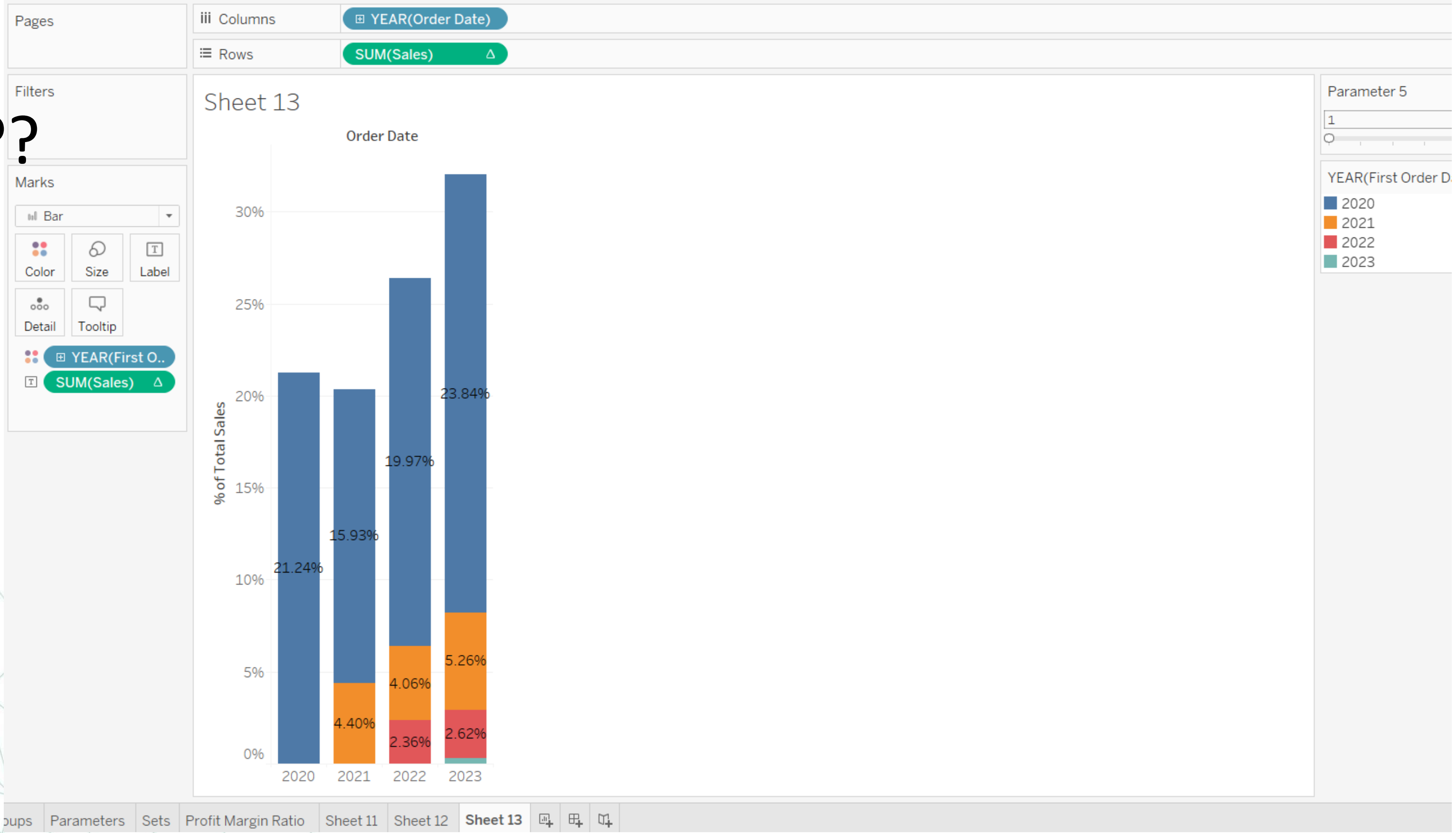
Practice Fixed LOD (10 Minute)

Visualization new and recurring customers

- Create a new worksheet.
- Drag Order Date to Columns (Year(Order Date)) and Sales to Rows.
- Change the visualization in the Marks card to Bar.
- Change SUM(Sales) to a Quick Table Calculation:
- Percent of Total → Compute Using Table (down)
- Create a Calculated Field and name it **First Order Date for Customer**
- { FIXED [Customer Name]: MIN([Order Date]) }
- Drag the newly created Calculated Field to Color
- Additionally, drag Sales to Label and change it to the same Table Calculation:
- Percent of Total → Compute Using Table (down)
- Now we can see the proportion of sales each year that stems from new versus recurring customers.

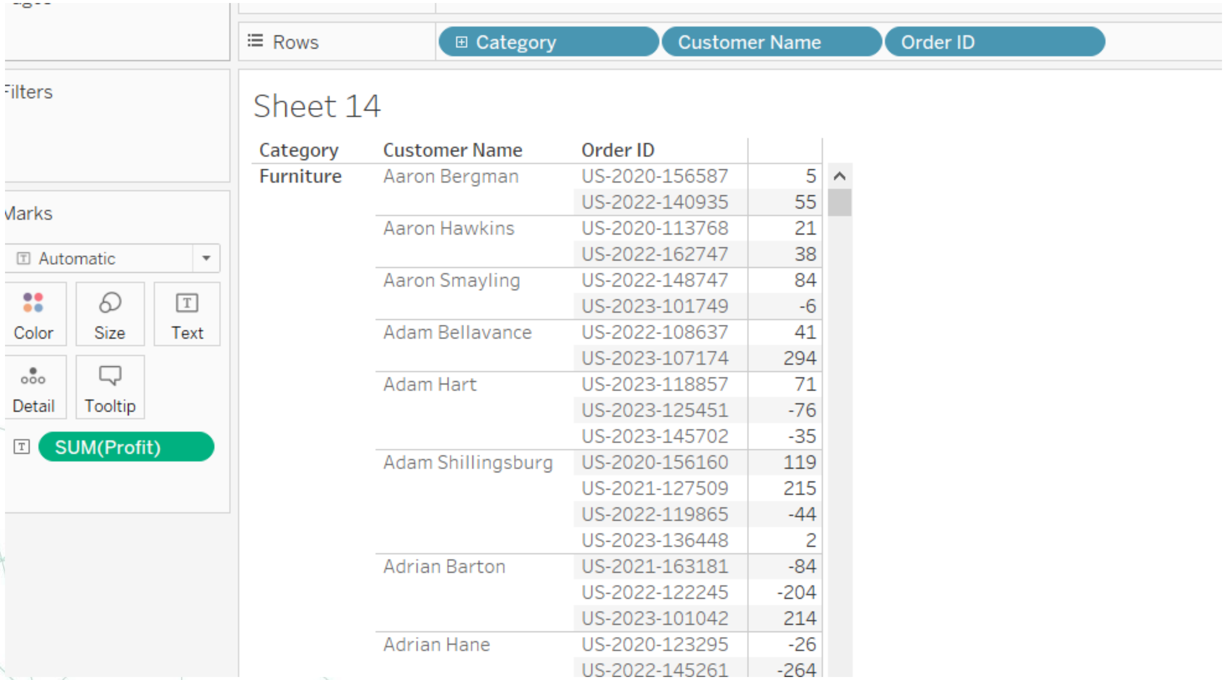
Practice Fixed LOD (10 Minute)

Interpret the result???



Include LOD

- It allows you to calculate at a finer level of detail in the database and then re-aggregate and show it at a coarser level of detail in your visualization.
- Drag Category, customer name and order Id to rows and Profit to label.
- What if we want the average profit per customer in each category



Category	Customer Name	Order ID	
Furniture	Aaron Bergman	US-2020-156587	5
		US-2022-140935	55
	Aaron Hawkins	US-2020-113768	21
		US-2022-162747	38
	Aaron Smayling	US-2022-148747	84
		US-2023-101749	-6
	Adam Bellavance	US-2022-108637	41
		US-2023-107174	294
	Adam Hart	US-2023-118857	71
		US-2023-125451	-76
		US-2023-145702	-35
	Adam Shillingsburg	US-2020-156160	119
		US-2021-127509	215
		US-2022-119865	-44
		US-2023-136448	2
	Adrian Barton	US-2021-163181	-84
		US-2022-122245	-204
		US-2023-101042	214
	Adrian Hane	US-2020-123295	-26
		US-2022-145261	-264

Include LOD

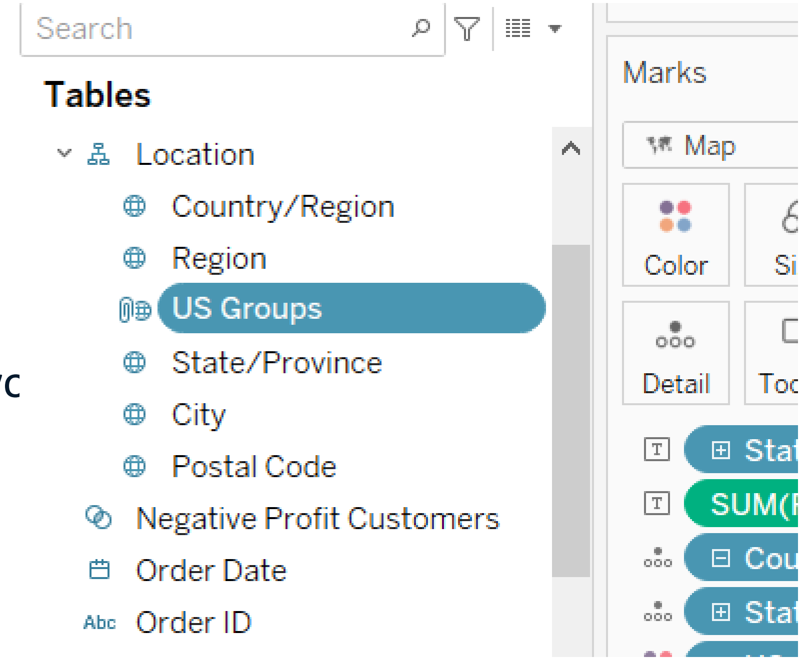
- Create the calculation field using Include.
- Rename it Avg Profit Per Customer
- Drag the category and the calculated field to rows
- Now we have avg profit per customer in each category.

Avg Profit per customer   Sample - Superstore

```
AVG({ INCLUDE [Customer Name]: SUM([Profit]) })
```

Exclude LOD Self Practice 10 minutes

- We remove a dimension from a worksheet
- Drag the previously created groups to the location hierarchy to be above state.
- create a calculated field name it 'area sales - exclude'
- use {exclude [State/Province]: sum([Sales])} formula
- add the newly created calculated field to text
- show how it changes over states and areas
- double check the numbers in another worksheet where you wc per areas



Let's practice

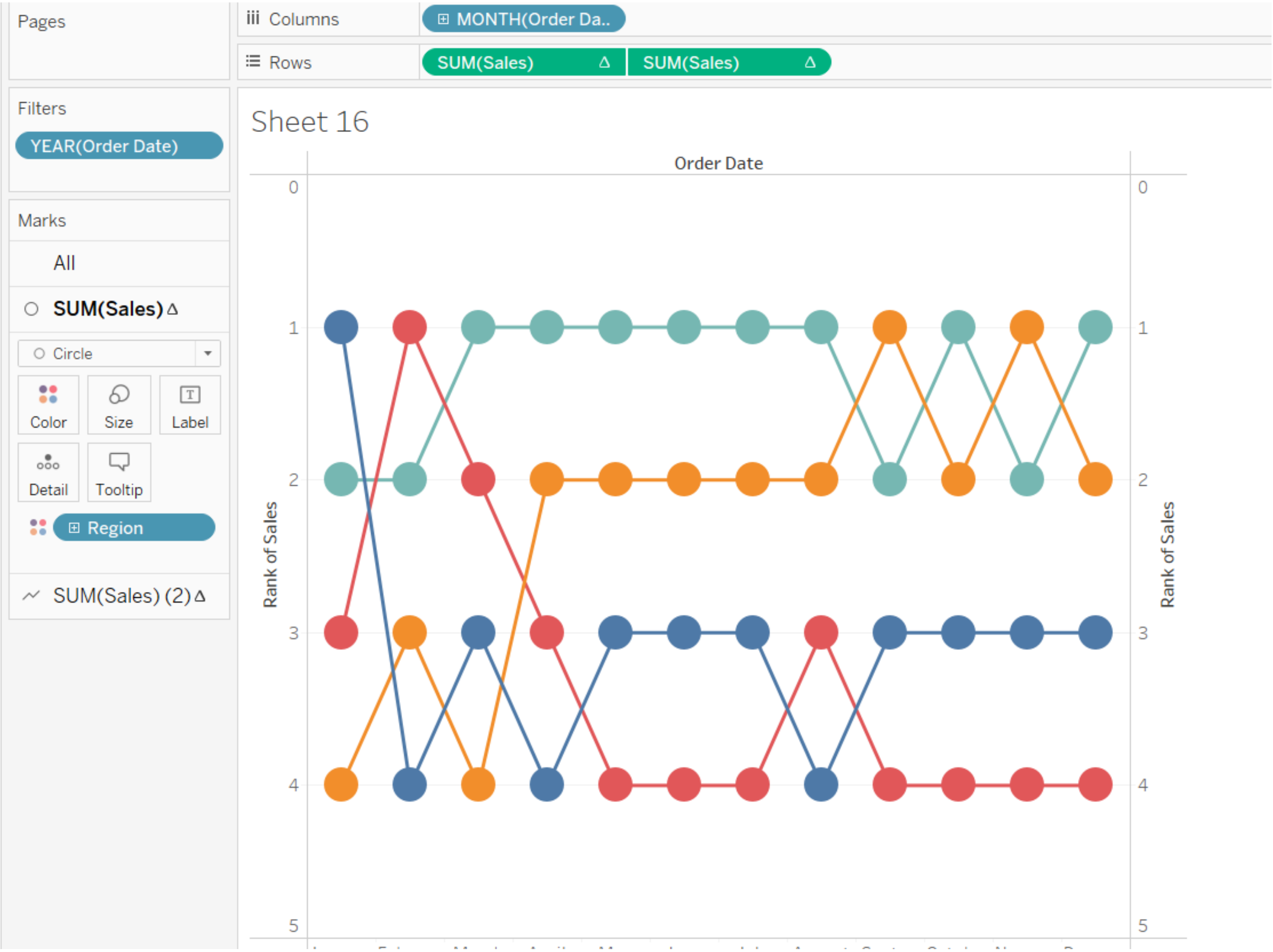
رواد مصر الرقمية

Self practice – 7 Minutes

وزارة الاتصالات وتكنولوجيا المعلومات

Bump Chart: https://www.numpyninja.com/post/bump-chart-in-tableau

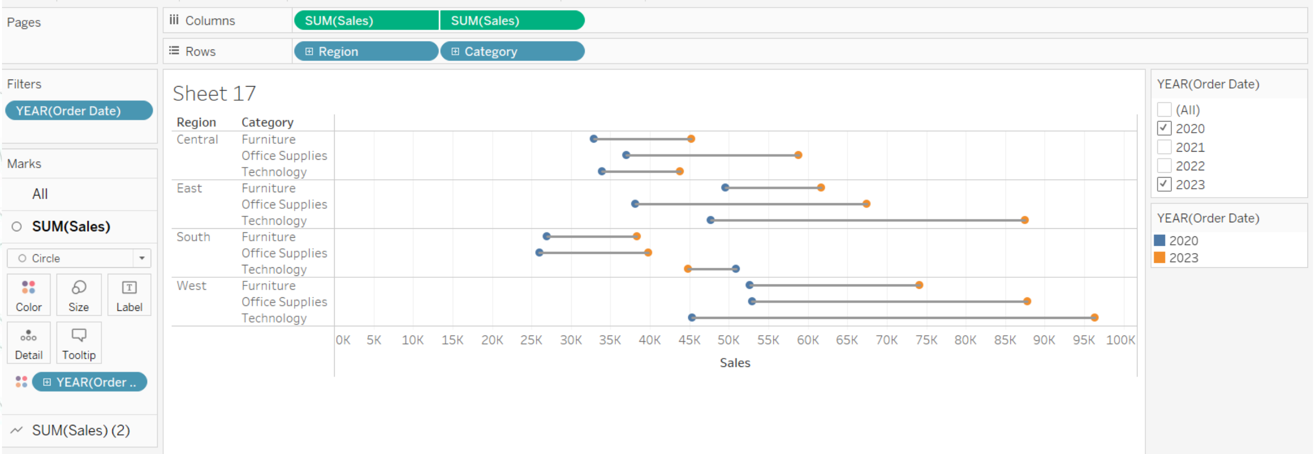
- Create a new worksheet
- Year-months to the columns
- Sales to the rows
- Region to color
- Now, we want sales to show the ranking of the region. For this, select Rank in the quick table calculation.
- Click again on the dropdown menu, compute using region
- Sales to rows again and add rank in the quick table calculation and compute using region.
- If you would like to have more granular display, filter one year and display for months (month in columns)
- Dual Axis then synchronize axis.
- Right click on axis and then reverse.



Self practice – 7 Minutes

Dumbell Charts

- create a new worksheet
- sales in columns
- region and category to the rows
- select line graph
- put order date into the path
- Drag the Year(Order Date) dimension to the filters shelf, select Years and click 2020(first year in the dataset) and 2023(last year in the dataset) -- sales in columns(again)
- Left click on SUM(Sales) and select dual axis
- Right click the top axis and deselect 'show header', then right click on the bottom axis to synchronise axis
- On the marks shelf change first chart type to a circle, drag Year(Order Date) to colours
- Interpret the results



Self practice – 5 Minutes

Customer Names and Data Protection:

- Create a new worksheet
- create a new calculated field
- to extract the first letter you can use left function: left([Customer Name], 1)
- extract the last name: split([Customer Name], ' ', -1)
- concatenate it together: left([Customer Name], 1) + ' ' + split([Customer Name], ' ', -1)
- update sales by customers visual with the newly created calculated field

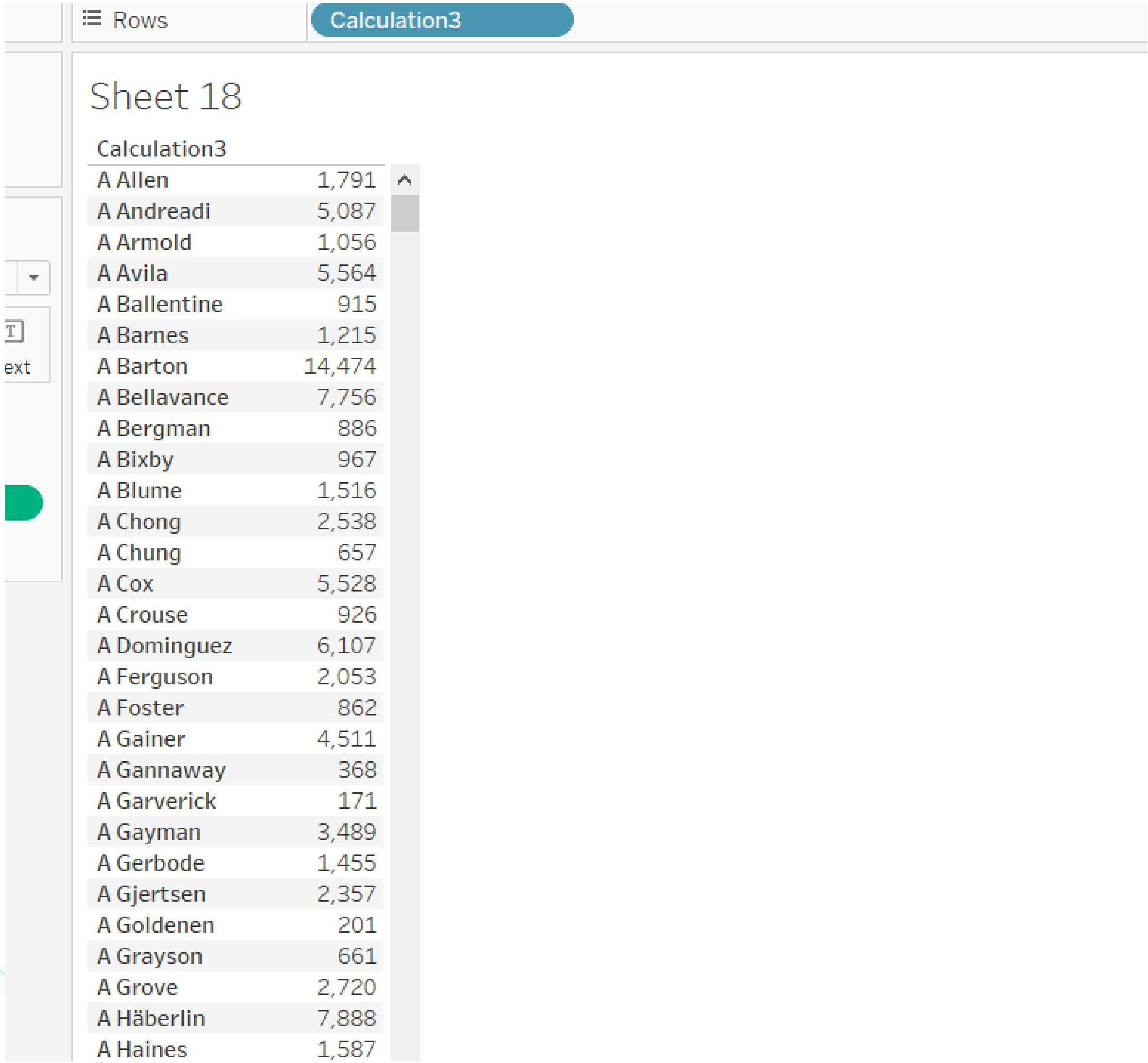


Table Calculations

- We want to calculate running totals:

- Drag year order date and month order date to columns
- Drag region to color
- Drag cnt(orders) to rows ,click on the dropdown menu and choose table calculation then running total , table across
- Repeat the last step in the label
- Choose bar chart

Interpret?

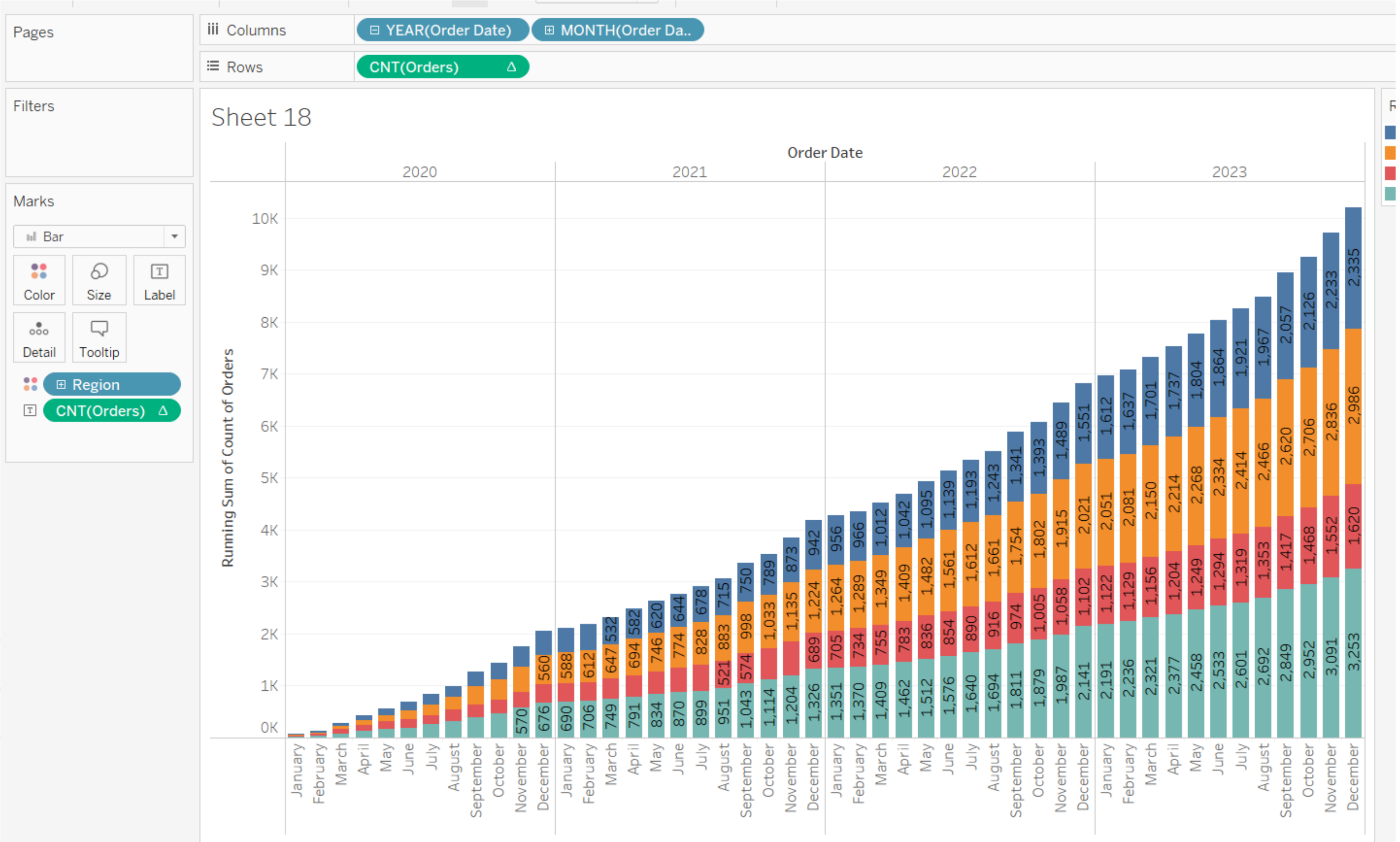


Table Calculations

- If we want to restart by the beginning of each year.
- All you have to do is change computing using and choose pane across
- Change it In both rows and marks pane

Interpret?

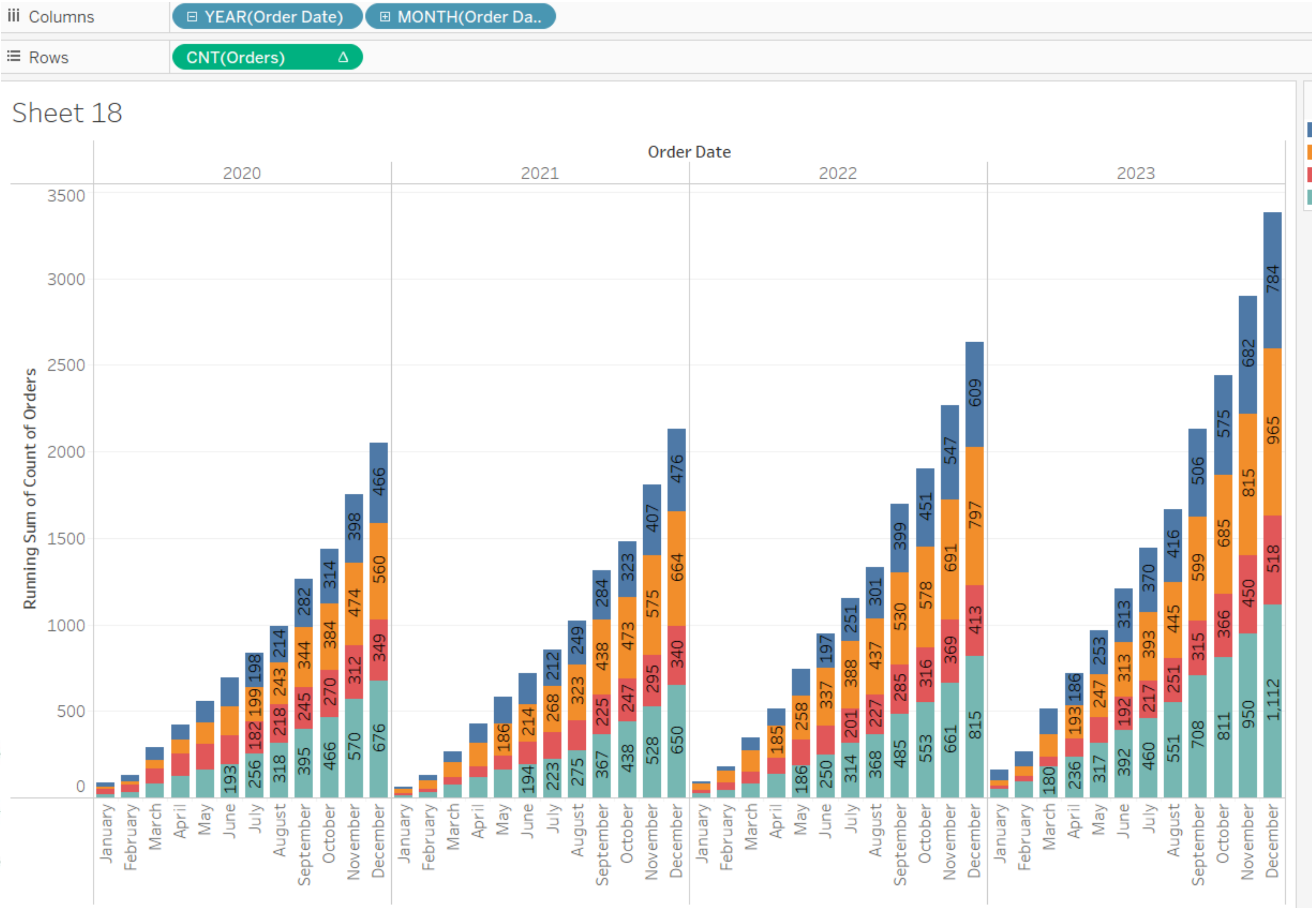


Table Calculations

- Move year order date to filters and choose only 2020, 2021
- Add the product category to the columns
- Note that the calculation is still performed table across.
- Remove Month and change back to table across
- Switch category with year order date

Interpret?

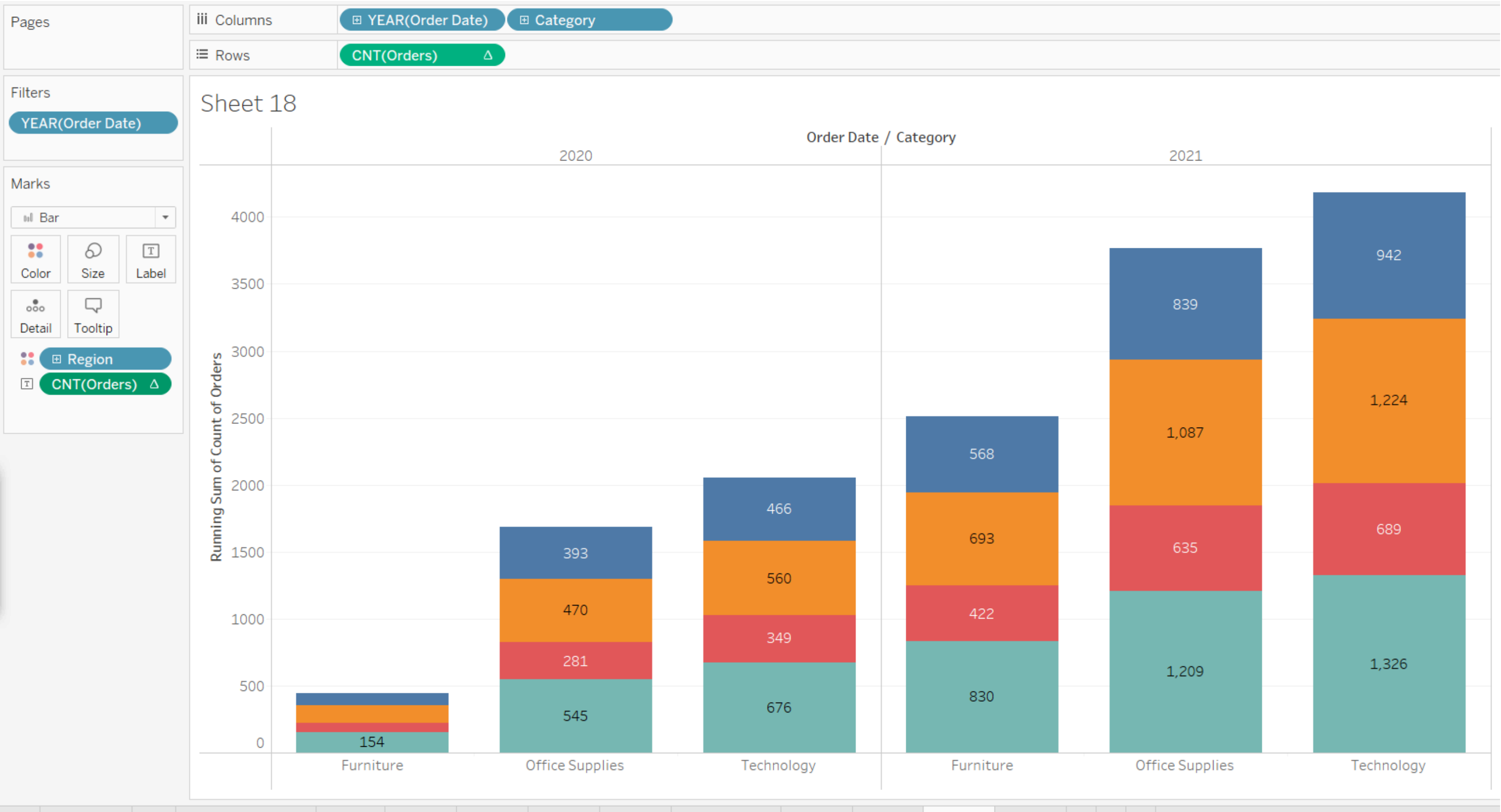


Table Calculations

- What will happen if we sorted the data in a different way?
- Sort the category ascending depending on the count of order id

Interpret?

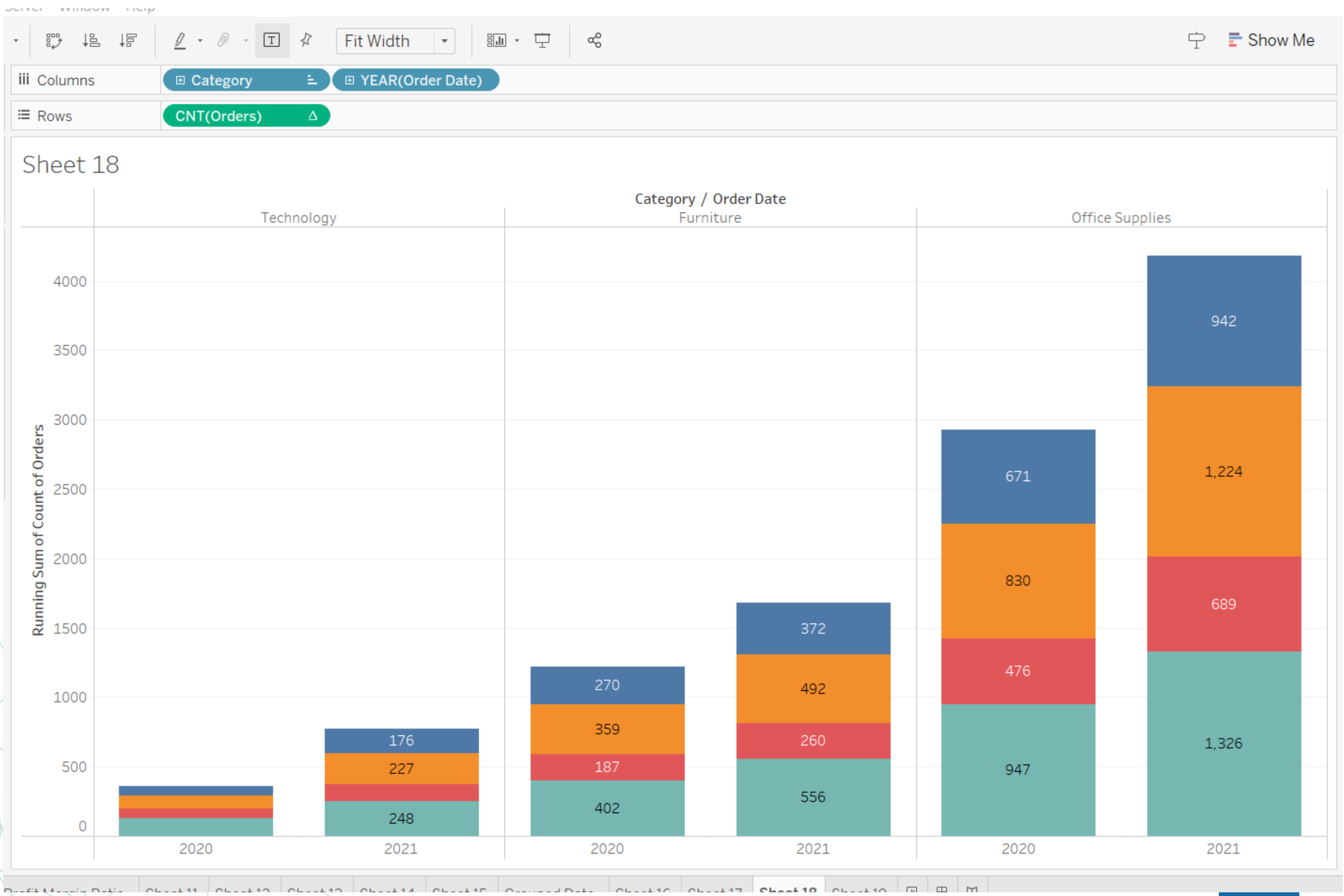
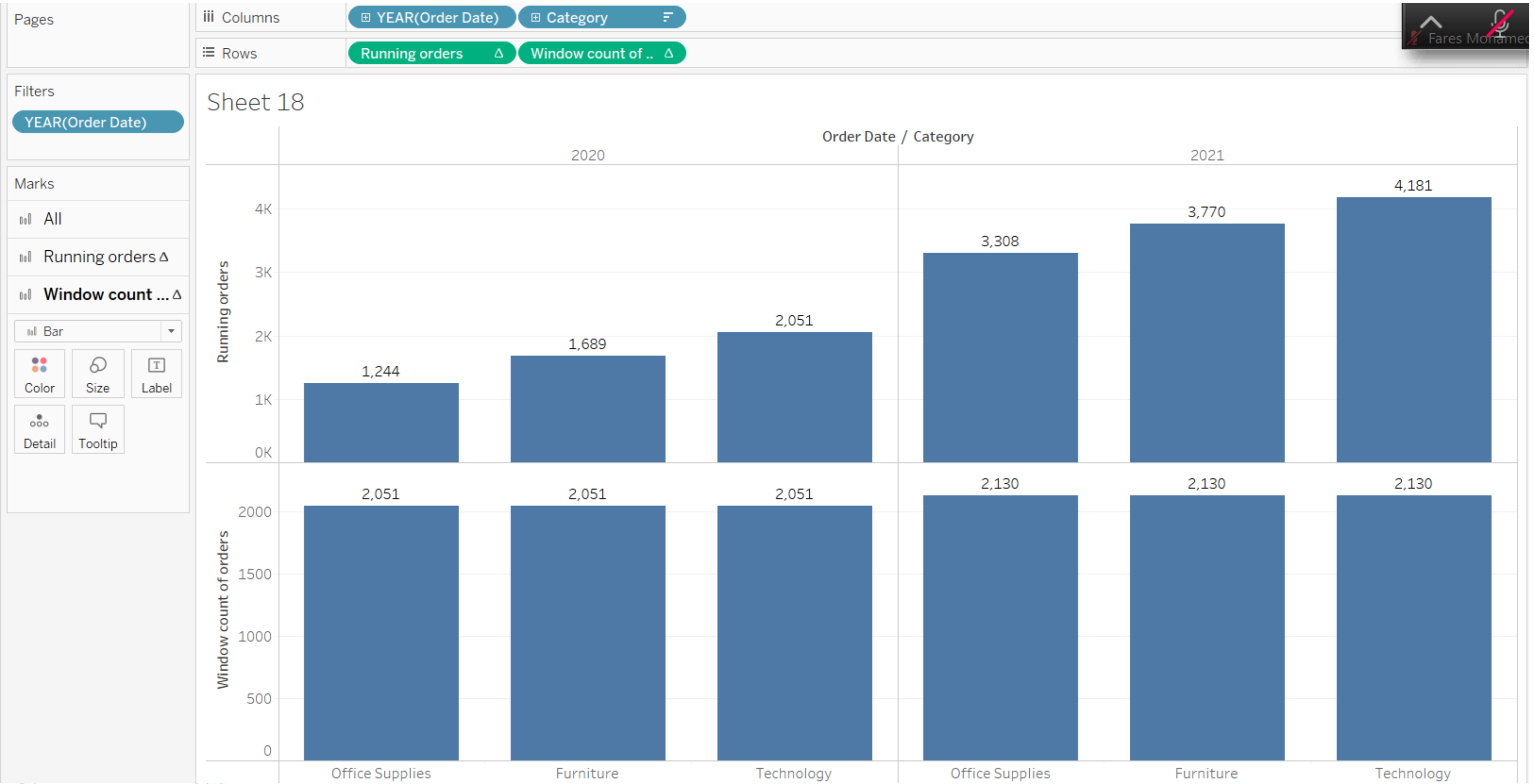


Table Calculations

- Create a new calculated field name it “Window Count of orders”
- WINDOW_SUM(COUNT([Orders]))
- Drag it to rows
- Compute using Pane across

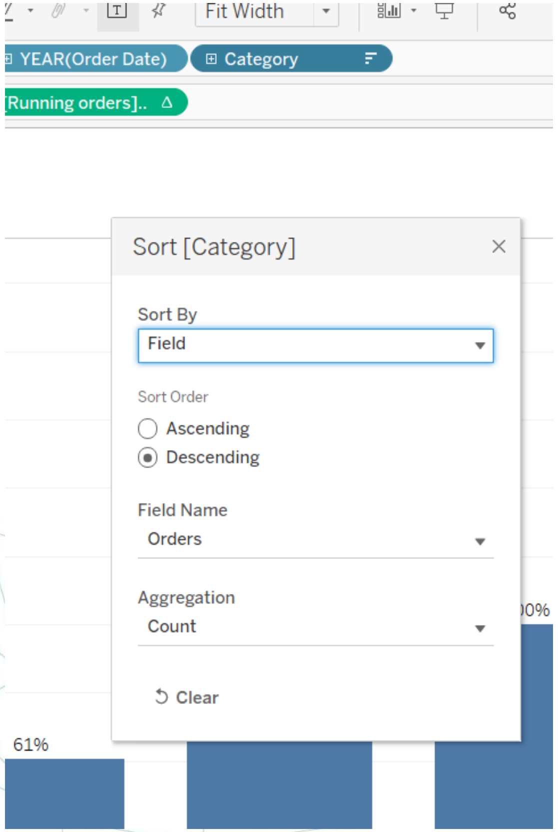


Interpret?

Table Calculations

- Click on an empty space in the row shelf and divide the two measures.
- Format > percentages
- Remove the other measures
- Sort category by cnt of orders desc.

Interpret?



Thank You